

4(a)(i)

description	level of protein structure
a gliadin protein is a single polypeptide that forms a compact structure	tertiary
20% of the amino acids in a glutenin molecule are glycine	primary
gliadin and glutenin molecules contain regions of β -pleated sheets	secondary

1 mark per correct row ;;;

3

Question	Answer	Marks
4(a)(ii)	<i>any one relevant:</i> e.g. RNA / gene, splicing, not needed / does not occur ; R DNA splicing <i>ref. to</i> all primary transcript is a coding sequence ; A primary transcript has the same sequence as the mRNA no nucleotides removed, from the primary transcript / after transcription ; no process of exon joining / AW ; no non-coding sequences, so introns to not need to be removed ; no alternative RNA splicing / exons not joined in different combinations ; A only one type of mRNA formed / mRNA formed always the same <i>idea that</i> only other post transcriptional processing occurs ; e.g. 5' G (guanine) cap / poly A tail	1
4(b)(i)	endocytosis ; A pinocytosis A described detail ; e.g. invagination of membrane / AW use of ATP A active process / energy needed formation of (endocytotic / pinocytotic) vesicle <i>if mp1 gained from description, mp2 must be another detail</i> <i>accept other named transport mechanism, with detail</i>	2

Question	Answer	Marks
4(b)(ii)	any four from: 1 ref. presence of, non-self / foreign, antigen or gliadin acts as antigen or ref. to antigen presentation (by macrophages) ; 2 recognition / activation / AW, of (specific) B- / T-lymphocytes ; A clonal selection 3 clonal expansion of B-lymphocytes (and T-lymphocytes) ; AW e.g. form a clone of cells (by mitosis) / divide many times by mitosis 4 T-helper cells, secrete / release, cytokine ; 5 cytokine stimulates, humoral / B-lymphocyte, response ; <i>if mp4 and mp5 not gained allow one mark for T-helper cells stimulate B-lymphocyte response</i> 6 B-lymphocytes (differentiate to) form plasma cells ; 7 plasma cells, produce / secrete, (anti-gliadin) antibodies ;	4